PURITY DESCENDS ALONG A MORPHISM FLAT AT THE POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Flatness of the base change of a morphism flat at a point, and lifting of specialisations along a flat morphism. Morphisms, label morphisms-lemma-base-change-flat (proof line 4967); Algebra, Section algebra-section-going-up (proof line 4970). The first makes the projection from the base change flat at the lifted point, and the second produces a specialisation upstairs lying over the given one.
- Points of a fibre product and base change of the relative assassin. Schemes, label schemes-lemma-points-fibre-product (proof line 4973); Divisors, label divisors-lemma-base-change-relative-assassin (proof line 4975). The first produces the required point of the fibre over the lifted specialisation and the second shows it is a relative associated point of the pulled back module.

Source reference. Original section 00HU: sources/stacks-chow/tags.txt, source lines 656–656. Navigation pointer for \ref{algebra-section-going-up} used inside the counted span (line 4970); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DC: sources/stacks-chow/tags.txt, source lines 6950–6950. Navigation pointer for \ref{divisors-lemma-base-change-relative-assassin} used inside the counted span (line 4975); the argument that consumes it is retained in the span.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Navigation pointer for \ref{morphisms-lemma-base-change-flat} used inside the counted span (line 4967); the argument that consumes it is retained in the span.

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Source reference. Original lemma 01JT: sources/stacks-chow/tags.txt, source lines 2026–2026. Navigation pointer for \ref{schemes-lemma-points-fibre-product} used inside the counted span (line 4973); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is of finite type. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $s \in S$. Let $(S', s') \to (S, s)$ be a morphism of pointed schemes. Assume $S' \to S$ is flat at s'.*

(1) *If $\mathcal{F}_{S'}$ is pure along $X_{s'}$, then $\mathcal{F}$ is pure along X_s.*
(2) *If $\mathcal{F}_{S'}$ is universally pure along $X_{s'}$, then $\mathcal{F}$ is universally pure along X_s.*

Proof. Let $(T \to S, t' \rightsquigarrow t, \xi)$ be an impurity of $\mathcal{F}$ above s. Set $T_1 = T \times_S S'$, and let t_1 be the unique point of T_1 mapping to t and s'. Since $T_1 \to T$ is flat at t_1, see Morphisms, Lemma 01U9, there exists a specialization $t_1' \rightsquigarrow t_1$ lying over $t' \rightsquigarrow t$, see Algebra, Section 00HU. Choose a point $\xi_1 \in X_{t_1'}$ which corresponds to a generic point of $\mathrm{Spec}(\kappa(t_1') \otimes_{\kappa(t')} \kappa(\xi))$, see Schemes, Lemma 01JT. By Divisors, Lemma 05DC we see that $\xi_1 \in \mathrm{Ass}_{X_{T_1}/T_1}(\mathcal{F}_{T_1})$. As the Zariski closure of $\{\xi_1\}$ in X_{T_1} maps into the Zariski closure of $\{\xi\}$ in X_T we conclude that this closure is disjoint from X_{t_1}. Hence $(T_1 \to S', t_1' \rightsquigarrow t_1, \xi_1)$ is an impurity of $\mathcal{F}_{S'}$ above s'. In other words we have proved the contrapositive to part (2) of the lemma. Finally, if $(T, t) \to (S, s)$ is an elementary étale neighbourhood, then $(T_1, t_1) \to (S', s')$ is an elementary étale neighbourhood too, and in this way we see that (1) holds. □